



COLLEGE CODE :9623

**COLLEGE NAME :AMRITA COLLEGE OF ENGINEERING AND
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ROLL NO autcs00055

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**COMPLETED THE PROJECT NAMED AS
PHASE 4 TECHNOLOGY PROJECT NAME:
INTERATIVE FORM VALIDATION**

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PROJECT TITLE: Invalid Form Validation

Phase 4 — Enhancements & Deployment

◇ 1. Additional Features

Added comprehensive form validation across all user input sections such as registration, login, and feedback forms.

Implemented real-time validation using JavaScript to instantly notify users about missing or incorrect data.

Introduced custom error messages to make instructions clear and user-friendly.

Integrated reset and auto-focus functions for error fields to enhance usability.

Enabled prevent-default mechanisms to stop form submission until all data is valid.

◇ 2. UI/UX Improvements

Improved form layouts with clear field labels, icons, and placeholder text for better user understanding.

Highlighted invalid fields with red borders and added tooltips showing specific validation messages.

Added success indicators (✓) when data is correctly entered, improving overall interaction flow.

Designed a mobile-responsive validation interface to ensure smooth performance across all devices.

Simplified the user experience by displaying inline error messages instead of pop-ups.

◇ 3. API Enhancements

Strengthened backend validation logic to double-check every input before saving to the database.

Implemented data sanitization and encoding to prevent malicious inputs or injections.

Modified API responses to include detailed error codes and messages for invalid submissions.

Integrated error-handling middleware to log invalid requests and assist in debugging.

Enhanced data consistency by validating dependencies between fields (e.g., password & confirm password).

◇ 4. Performance & Security Checks

Conducted load testing to ensure validation scripts do not affect form performance.

Verified that all inputs meet predefined constraints (like length, format, and type).

Implemented rate limiting and CAPTCHA to prevent spam submissions.

Secured sensitive inputs like passwords using encryption and hashing techniques.

Ensured compliance with security best practices (OWASP validation rules).

◇ 5. Testing & Deployment

Performed unit and integration testing for validation functions using tools like Jest and Postman.

Tested edge cases such as empty fields, wrong email formats, and invalid characters.

Collaborated with the QA team for user acceptance testing (UAT) to verify usability.

Fixed all identified issues before deployment.

Deployed the final validated project on Netlify/Vercel/Cloud platform for public access.

Verified that form validation works perfectly after deployment across browsers and devices.